

Retail Sales Forecasting & Customer Purchase Analysis

Final Data Science Project Report

Project Area	Retail sales analysis and forecasting
Tools	Python, Pandas, NumPy, Matplotlib/Seaborn, Scikit-learn, Power BI
Forecasting Method	Linear Regression
Forecast Horizon	Next three months

Objective: analyze historical retail sales, identify meaningful sales patterns, evaluate a predictive model, forecast the next three months, and present business-oriented insights.

1. Project Overview

The project analyzes historical retail transaction data to understand revenue performance across product categories, subcategories, and sales regions. A time-based Linear Regression model was then used to produce a three-month revenue forecast. The results were presented through a Power BI dashboard.

2. Data Collection & Preprocessing

The retail dataset was imported and processed in Python using Pandas. The preprocessing workflow included handling missing values, removing duplicate records, converting date fields into datetime format, and creating analysis-ready time features.

- Missing values: missing/unknown values were handled during the cleaning stage.
- Duplicates: duplicate records were removed.
- Date processing: the order-date field was converted to datetime format.
- Feature engineering: Month, Quarter, Weekday, YearMonth, and Total Revenue were created/used for analysis.
- Revenue: Total Revenue was used as the main business target for aggregation and forecasting.

Dataset field limitation

The supplied dataset does not contain a direct Customer Segment field. It also does not contain a direct Store Location field; SalesRegion was therefore used for geographic performance analysis. These substitutions are documented rather than presented as original fields.

3. Exploratory Data Analysis

The analysis examined revenue over time and across product and geographic dimensions. The main visual analyses included monthly sales trend, product category performance, sales-region performance, product subcategory performance, and correlation analysis.

Analysis	Observed result
Category revenue	Bikes generated the highest category revenue in the Python analysis (~33.64M).
Top subcategories	Road Bikes (~14.33M) and Mountain Bikes (~13.96M) were the largest subcategory contributors.
Regional revenue	Southwest was the highest-revenue sales region in the Python aggregation (~9.52M).
Monthly pattern	Revenue fluctuated across the available monthly observations rather than following a perfect trend.
Correlation	Unit Price showed a correlation of ~0.73 with Sales Amount; List Price was ~0.73 as well.

4. Predictive Modeling

A Linear Regression model was used to model monthly revenue against a time index. The historical monthly data was divided into training and testing portions. The model was trained on the training period, evaluated on the held-out test period, and then refitted on the available historical observations to generate the next three monthly forecasts.

Metric	Result
Mean Absolute Error (MAE)	761,855.59
Mean Squared Error (MSE)	888,402,908,319.06

Metric	Result
Root Mean Squared Error (RMSE)	942,551.28
R ² Score	-0.1877

The negative R² indicates that this simple time-based Linear Regression model did not explain the held-out revenue variation well. The result should therefore be treated as a baseline model rather than a high-accuracy forecasting system. More observations and models that incorporate seasonality, product mix, discounts, region, and other explanatory variables could be evaluated in a future iteration.

5. Three-Month Sales Forecast

Month	Forecasted Revenue
July 2013	3,016,058
August 2013	3,046,226
September 2013	3,076,394

The model forecasts a gradual increase from approximately 3.02M in July to approximately 3.08M in September 2013. Because model evaluation produced a negative R², these values should be used as model-generated baseline estimates rather than precise commitments.

6. Business Insights & Recommendations

- Category focus: Bikes are the largest revenue contributor in the analyzed Python aggregation. Inventory and availability planning can therefore pay particular attention to the bike category.
- Product planning: Road Bikes and Mountain Bikes are major subcategory contributors, so stock levels, supplier lead times, and replenishment planning can be monitored closely for these products.
- Regional planning: Revenue differs substantially across sales regions. Inventory and staffing can be reviewed region-by-region rather than using one uniform allocation.
- Seasonality: Monthly revenue fluctuates across the observed period. Promotional calendars and inventory replenishment can be aligned with historically stronger and weaker months.
- Promotions: Customer-segment-specific recommendations cannot be made from this dataset because a Customer Segment field is not available. Promotion analysis can instead be based on product/category and regional performance.
- Sales drivers: Unit Price and List Price show relatively strong positive correlations with Sales Amount in the calculated correlation output. Correlation should not be interpreted as proof of causation.

7. Power BI Dashboard

The dashboard combines KPI cards, historical trend analysis, category and regional comparisons, subcategory performance, a three-month forecast, and a concise key-insights section.

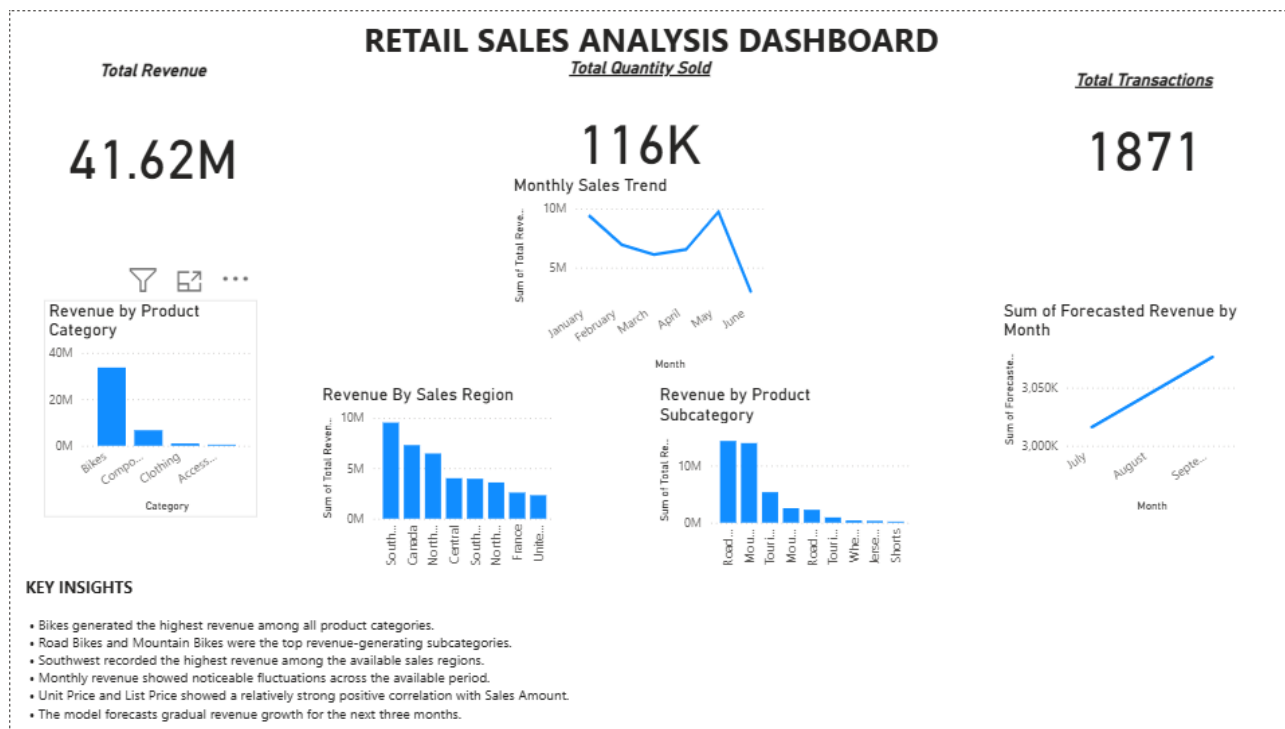


Figure 1. Power BI dashboard screenshot supplied for the project submission.

Dashboard data consistency note

Before final submission, compare the KPI cards with the underlying visuals. The supplied screenshot shows a Total Revenue card of 6.54M while the category/regional visuals and the Python aggregation contain substantially larger totals. This may indicate a Power BI filter, relationship, aggregation, or data-loading issue. The dashboard should be checked so all visuals use the same cleaned dataset and the same date/filter context.

8. Conclusion

This project demonstrates an end-to-end retail analytics workflow: data cleaning, feature engineering, exploratory analysis, predictive modeling, forecasting, and dashboard reporting. The analysis identifies major product and regional revenue patterns, while the Linear Regression model provides a simple baseline for short-term forecasting. The model's evaluation results also show why forecast outputs should be interpreted cautiously and why richer forecasting approaches could be considered for future work.

9. Technology Stack

Python: Pandas, NumPy, Matplotlib/Seaborn, Scikit-learn Visualization: Power BI Report: PDF